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REVIEW ARTICLE

Mutations Associated with Pyrazinamide Resistance in *Mycobacterium tuberculosis*: A Review and Update

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Abstract
Pyrazinamide (PZA) has remained a keystone of tuberculosis (TB) therapy, and it possesses high imperative sterilizing action that can facilitate reduction in the present chemotherapy regimen. The combination of PZA works both with first- and second-line TB drugs, notably fluoroquinolones, clofazimine, bedaquiline, delamanid and pretomanid. Pyrazinamide inhibits various targets that are involved in different cellular processes like energy production (*pncA*), trans-translation (*rpsA*) and pantothenate/coenzyme A (*panD*) which are required for persistence of the pathogen. It is well known that *pncA* gene encoding pyrazinamidase is involved in the transition of PZA into the active form of pyrazinoic acid, which implies that mutation in the *pncA* gene can develop PZA resistance in *Mycobacterium tuberculosis* (*M. tuberculosis*) strain leading to a major clinical and public health concern. Therefore, it is very crucial to understand its resistance mechanism and to detect it precisely to help in the management of the disease. Scope of this review is to have a deep understanding of molecular mechanism of PZA resistance with its multiple targets which would help study the association of mutations and its resistance in *M. tuberculosis.* This will in turn help learn about the resistance of PZA and develop more accurate molecular diagnostic tool for drug-resistant TB in future TB therapy.

Introduction

Mycobacterium tuberculosis (*M. tuberculosis*) is the causative agent of tuberculosis (TB), which is a chronic infectious disease with high fatality rate [1]. In 2019, globally 10 million people were affected with TB with 7.1 million new cases and 1.4 million succumbed to death. In the context of cumulative reduction of TB cases, prevalence of drug-resistant TB happens to be a serious complication in community health. Approximately 5,00,000 people advanced to rifampicin-resistant TB (RR-TB), and among which 78% had multidrug-resistant TB (MDR-TB). The prevalence of MDR-TB could be found largely in the Asia Pacific region which includes India (27%) followed by China (14%) and Russia (8%) [1]. This emphasises the relevance of studying drug resistance and its associated mutations. Pyrazinamide (PZA) resistance has eventually turned to be ubiquitous, with approximately one out of six incidents of TB cases having PZA resistance and the evaluated global burden of fresh PZA resistant TB cases is 1.4 million per year which includes 2,70,000 MDR-TB cases with PZA resistance [2]. The occurrence of PZA resistance has a strong correlation with resistance to rifampicin, and it has been observed that it elevates beside MDR and extremely drug resistant TB (XDR-TB) strains. It is important to find PZA resistance by performing appropriate drug susceptibility tests (DST) [3]. Inclusion of PZA in the TB treatment regimen can decrease the treatment duration from 9 to 6 months in drug susceptible TB patients, and its' destructive property towards persistent tubercle bacilli in the course of intensive phase of chemotherapy helps in enhancing the outcome of clinical TB treatment for non-MDR and MDR-TB. At present, the therapeutic effect of PZA with bedaquiline, delamanid, PA-824 and moxifloxacin is being explored in phase 3 clinical trials [4]. Other clinical trials involving PZA are MDR-END TB (9–12-month regimen with delamanid, linezolid, levofloxacin and PZA for the treatment of MDR-TB patients with no resistance to fluoroquinolones—Republic of Korea), TRUNCATE—TB (Phase II/III trial—2 month regimen with isoniazid, PZA, ethambutol, linezolid and rifampicin; isoniazid, PZA, linezolid, rifapentine and levofloxacin; or isoniazid, PZA, ethambutol, linezolid and bedaquiline—Indonesia, the

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Philippines, Singapore and Thailand) and NeXT (Phase III trial. Evaluation of a 6−9-month regimen of bedaquiline, ethionamide or high-dose isoniazid, linezolid, levofloxacin and PZA for MDR-TB patients—South Africa) [5].

In the current context of drug administration, there is a need for deeper insight of PZA molecular mechanisms of action and resistance in *M. tuberculosis* because of its inevitable role in the present and future drug regimen for both drug-susceptible and drug-resistant TB [6]. The definite mechanism of PZA is obscure because its action depends on acidic environment of the host macrophage, and unreliable results of PZA susceptibility tests using acidic medium complicates the research on its mechanism of action. It also leads to inappropriate treatment of TB followed by spread of PZA resistance [7]. While BACTEC-MGIT 960, a drug susceptibility test, is regularly done to assess the first line and second line drugs resistance, PZA has just begun to be used during clinical testing. However, there is a risk of getting ambiguous outcomes. Therefore, only whole genome sequencing could be done to have findings on the mutations involved in PZA resistance as it is associated with mutations in various genes; as it contributes in high-risk frequency due to its co-resistance to other drugs, this procedure will help in the management and control of MDR-TB epidemic. In 2015, a systematic review summarized the data and showed the incidence of PZA to be rising as global public ill health and prevalence of resistance to PZA [2]. In another study, Ramirez-Busby and Valafar et al. characterized the mutations in pyrazinamidase (PZase) and *pncA* promoter and further assessed the endurance of its association between mutations and phenotypic resistance to PZA, however did not review the unknown mechanism of the targets [8]. This review is an attempt to update and consolidate evidences from the studies published since 2000, on the relevance of mutations and resistance to PZA, also, to further understand the mechanism of targets which is a step towards the importance of early and easy method to identify PZA resistance.

Discovery of PZA

Pyrazine-2-carboxamide (C5H5N3O—M.W. 123.11) is a synthetic amide derivative of pyrazinoic acid. PZA is structurally similar to nicotinamide which was first chemically synthesized in 1936 [9]. In 1945, Vital Chorine discovered that nicotinamide has certain activity against mycobacteria in guinea pigs and mice. While treating *Mycobacterium leprae* infected rats and mice with high concentrations of nicotinamide, Chorine found that it prolonged their survival [10]. During early 1950s concern regarding PZA as an anti-TB drug arose, because of its rapid drug resistance and potential to cause hepatotoxicity [11]. In 1952, Yeager et al. reported that PZA had higher activity in mice and based on the unexpected findings of nicotinamide, it has been recognized as an anti-tuberculosis drug [12]. The knowledge of the nicotinamide analogues also helped in the invention of the effective anti-tuberculosis drugs isoniazid (INH) and ethionamide (ETH) [13]. Difficulty in understanding of PZA mechanism of action causes emergence of drug resistance and depreciated results in *in vitro* conditions has been a roadblock for PZA's global recognition, but due to its performance in *in vivo* conditions, PZA was recommended as one of the most important drug in the chemotherapy of tuberculosis along with isoniazid and rifampicin in the 1980s [14]. Post 1958, PZA was introduced as a second line drug to treat chronic resistant cases of isoniazid and streptomycin. Development in clinical trials of short-course treatment of new cases from 1971 to 1980 confirmed PZA's vital role during TB treatment [11].

Deciphering Mutations in Pyrazinamide Resistance

Resistance to PZA develops due to the presence of various mutations in multiple targets that includes *pncA, rpsA, panD*, c*lpC1* and other unidentified mechanism of genes. Mutations in the target genes affect flexibility and deviation and make them a weak target for interaction with drugs. Each target is involved in different mechanism of action which is represented by a model (Fig. 1). Pyrazinamide resistance is primarily associated with mutation in *pncA*

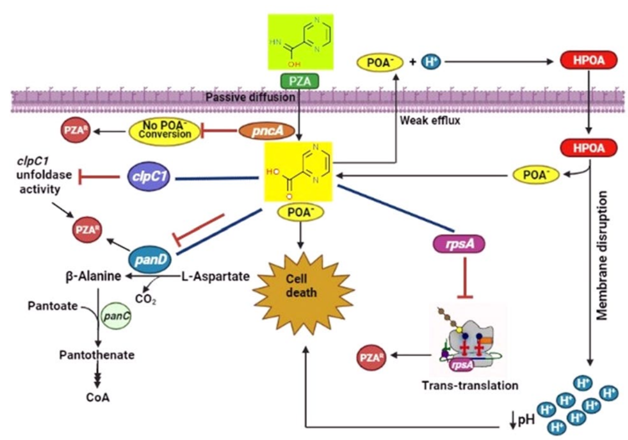

Fig. 1 Mechanism of action on Pyrazinamide drug targets in susceptible and resistant bacilli. PZA enters the cell by diffusion and is activated by *pncA* and converted into POA. POA acts as a protonophore causing acidification of the mycobacterial cytoplasm (*pncA*) causing cell death. Whereas inhibition of POA conversion, inhibition of trans-translation (*rpsA*), inhibition of pantothenate pathway (*panD*) and inhibition of unfoldase activity (*clpC1*) causes PZA^R. *PZA* pyrazinamide, PZA^R pyrazinamide resistance, POA^- pyrazinoic acid; ⊣—denotes inhibition.

and its promoter region according to recently released WHO mutation catalog in 2021 [1].

PncA Action

pncA gene has 561 nucleotides which is present in *M. tuberculosis* that encodes PZase, helps in the conversion of the pro drug PZA into its active form pyrazinoic acid (POA) [15]. The mode of action of PZA passes through the mycobacterial cell membrane by passive diffusion where it is transformed into POA in the cytoplasm and is released by a weak efflux pump into the acidic circumstances [16]. The acidic POA forms uncharged HPOA through protonation, which is easily absorbed by the cell and intracellular dissociation takes place; this results in the release of H^+ protons into the cytoplasm. The continuous release of H^+ can cause cellular damage and cell death due to intracellular acidification [17]. In addition, protonated form of POA (HPOA) has the potential to destroy the membrane by disintegrating the proton motive force resulting in disruption of membrane transport [11]. In in vitro conditions, besides acidic pH, other factors that can affect the activity of PZA are age of the culture, inoculum size, levels of serum albumin, iron content, UV, weak acids and hypoxic conditions [18, 19]. Drug exposure assays are preferred over conventional minimum inhibitory concentration (MIC) to better highlight the factors influencing the action of PZA. This is due to its strange nature as its action is elevated towards non-growing tubercle bacilli when compared to that of growing bacilli. This property is contradictory when compared to other familiar antibiotics [18].

pncA Gene Mutation

A mile stone chart is created for the PZA resistance associated with *pncA* mutations to visualize the significant developments and findings in *pncA* gene mutations from the 1967 to 2020 studies (Fig. 2). Pyrazinamide resistance in *M. tuberculosis* is associated with loss of nicotinamidase and PZase as shown by McDermott's group in 1967 [16]. The nature of *pncA* mutations that contribute to the primary mechanism of PZA resistance has missense mutations that cause amino acid substitutions (Asp63 (GAC) → His (CAC), Cys138 (TGT) → Ser (AGT), Gln141 (CAG) → Pro (CCG). Mutation in the *pncA* gene is known to cause PZA resistance and showed an indisputable correlation between PZA resistance and loss of enzyme activity [15]. Transformation of *pncA* gene from *M. tuberculosis* into PZA resistant BCG led to the determination of genetic basis of PZA resistance and it was found that His57Asp single mutation in PZase was responsible for PZA resistance in *M. bovis.* Therefore, *pncA* of *M. tuberculosis* conferred PZase activity and also PZA susceptibility in *M. bovis* [20]. Sreevatsan et al. also validated these findings using automated DNA sequencing (IS6110 typing) and demonstrated that *pncA* mutations are

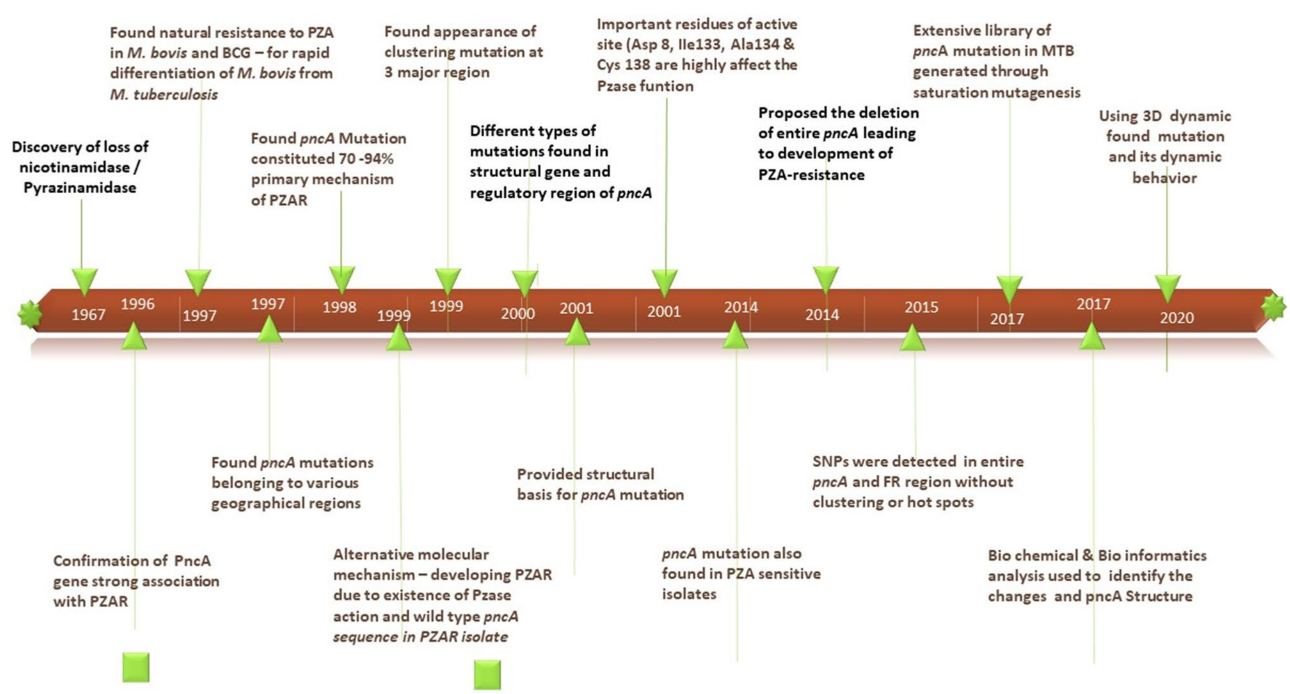

Fig. 2 Mile stone chart for *pncA* mutation association of PZA resistance. The mile stone chart depicts the PZA resistance associated with *pncA* mutations to help visualize the significant developments and findings in *pncA* gene mutations from the 1967 to 2020 studies

related to PZA resistance, despite belonging to various geographical regions and are dispersed throughout the gene, including the 82 bp open reading frame putative regulatory region. The loss of PZase action and the mutants' inability to activate PZA to form POA can be attributed to these mutations [21]. In certain cases, different types of mutations like nucleotide insertions or deletions, nonsense mutations in the structural gene, or in the promoter and regulatory region of *pncA* are also found to be responsible [22, 23]. Several studies have indicated the occurrence of alternate molecular mechanism for developing PZA resistance which is due to the existence of PZase action and wild-type *pncA* sequences in PZA-resistant isolate, though 70–97% PZA resistant contain mutations in their putative regulatory region or *pncA* gene [11].

High diversity of *pncA* mutations were observed in *M. tuberculosis* complex around various geographical regions [24]. This may be due to adaptive mutagenesis or the inability during DNA mismatch repair in *M. tuberculosis* [25], also due to *pncA* gene location. However, certain degree of mutations in the *pncA* gene occur to cluster within three major regions:—3–17, 61–85, 132–142. These hold catalytic sites for the PZase enzyme. These catalytic residues comprise the active site (D8, K96, A134 and C138) including the metal-binding site (D49, H51 and H71) [26]. Mutations which produce physical–chemical modification of the active site or the metal-binding site cause significant loss of PZase activity [27]. Muthaiah et al. claimed that 79% mutations were confined to three regions of *pncA* in *M. tuberculosis;* it was also found that two isolates contained 5.1% unique mutation at amino acid 26 (Ala → Gly), and *pncA* mutations were dispersed along the gene [28].

The 70% of regions that were affected the most in mutated strains were at the promoter and at codons 6–15, 50–70, 90–100, 130–145 and 170–175. Mutation in the regulatory region (A-11G) of the *pncA* gene is very customary [29]. In this regard, another study showed 94.4% of *pncA* mutations were amino acid substitutions, insertions or deletions of nucleotides which resulted in nonsense mutations in the *pncA* promoter region [30]. Single nucleotide polymorphisms (SNP) have been reported throughout the entire *pncA* gene and flanking region without specific clusters or hot spots and few polymorphisms in *pncA* were also reported in susceptible isolates [8, 29]. The extensive library of *pncA* mutations in *M. tuberculosis* was generated by saturating mutagenesis and observing more than 300 substitutions in vitro and in vivo that confer PZA resistance. The in vivo model suggested that the unique host-mediated selection pressure is not captured by in vitro experimentation. In addition, many non-synonymous substitutions were also reported that did not cause PZA resistance [31]. The biochemical and bioinformatics analysis indicated that most PZA resistance conferring substitutions caused changes in the *pncA* structure and stability or disrupted active site catalytic triad and iron coordinating residue [7]. Further, a structural analysis revealed that a few precise mutations—A46V, H71Y and D129N—in PZase may be involved in changing its dynamic behavior, leading to weak interaction with PZA and inducing resistance to PZA [32]. Pang et al. detected that a frequent mutation in the *pncA* gene was the nucleotide substitution from A to G at the 11th position in the promoter region of *pncA*, which alone accounted for 5.4% of mutant isolates, possessing negative PZase activity in PZA-resistant strains. But Hameed et al. identified that it leads to positive PZase activity when C nucleotide at 114 position is deleted of upstream flanking region (UFR) of *pncA*. This alteration induced normal regulation of PZase (PZase positive) in this PZA resistance strain [33].

In a 2015 study, the prevalence of PZA resistance was reported globally in all six WHO defined regions (Africa, Americas, Eastern Mediterranean, Europe, South East Asia and Western Pacific). The pooled PZA resistance prevalence is estimated to be 16.2% among all TB cases, 41.3% among patients at high risk of MDR-TB, and 60.5% among MDR-TB cases. The global burden is 1.4 million new PZA-resistant TB cases annually. In all six WHO regions, there was a two to sixfold increase in PZA resistance among MDR-TB patients compared to the population of all TB patients [34].

Studies on DNA sequencing have shown that the *pncA* gene is mutated along its whole length, indicating that sequencing the complete *pncA* gene would be necessary to detect all probable mutations. A 2015 meta-analysis found that 61 studies revealed a total of 641 mutations in 171 (of 187) codons from 2760 resistant strains and 96 mutations from 3329 susceptible strains [8].

In this review, we find that the frequency of mutations in the *pncA* gene in PZA-resistant *M. tuberculosis* strains ranging from 40 to 90% in different studies conducted in different populations is listed in Table 1 which includes data from 2000 to 2021, covering regions that are considered as hot spots—Japan, China, South Africa, India and other low prevalence regions. The studies selected for this table were obtained by entering keywords 'PZA resistance', '*pncA* mutation', 'PZA associated mutation', '*rpsA* mutation', '*panD* mutation', '*clpC1* mutation', '*mas* and *ppsA-E* mutation' in the Google and PubMed search engine. The frequency of PZA resistance based on *pncA*, *rpsA* and *panD* mutations are mentioned in Table 1.

RpsA Action

The ribosomal protein S1 (RpsA) is an essential translation protein, which binds to ribosome and upstream sequences of mRNA. The C-terminus of the RpsA specifically binds to tmRNA, which is a major step in Trans-translation, which

Table 1 Frequency of PZA resistance based on *pncA*, *rpsA*, *panD and clpC1* mutation

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
1	Hou et al. (2000) [76]	China	35 PZA-resistant isolates and 30 PZA-susceptible isolates were identified. 32 out of 35 PZA-resistant isolates had *pncA* mutation	Mutations were found dispersed along the entire *pncA* gene 22 isolates had Point mutation causing amino acid substitution 6 isolates had nucleotide deletion and 4 isolates had nucleotide insertion frameshifts mutation Previously uncharacterized mutations were also found which cause the following changes: Asn118 → Thr, CG insertion at position 501; CC insertion at nucleotide position 403; an 8 base-pair deletion at start codon; Pro54 → Thr; AG insertion at 368; Tyr41 → His, Ser88 → stop, and A insertion at nucleotide position 301	Targeted sequencing and SSCP	*pncA* 91.4%
2	Lee et al. (2001) [77]	Korea	A total of 95 PZA-resistant clinical isolates; 92 (97%) exhibited *pncA* mutations	Mutations were found in 73 variations at 71 dispersed sites throughout the *pncA* gene including the upstream region and 13 variations were seen in more than two isolates 72 isolates had nucleotide substitutions and 10 had deletions and 10 had insertions Three different types of nucleotide substitutions were observed First, 6 isolates carried mutations at -11 (A → G), the upstream region of the gene Second, 6 isolates had single nucleotide substitutions at positions 28, 203, 309, 357 or 421 presumably leading to premature termination of protein synthesis Third, 60 isolates had amino acid replacement Overall, 44 mutations were newly identified in this study	Targeted sequencing	*pncA* 97%
3	Bishop et al. (2001) [78]	South Africa	A total of 15 PZA resistant clinical isolates; 10 had *pncA* mutations. Six isolates were PZA susceptible	Mutations were found dispersed along the entire *pncA* gene Nucleotide substitutions were identified in 7 isolates, insertions in 2 isolates and deletions in 1 isolate causing amino acid substitutions Mutation in codon 76 (Thr-Pro) and amino acid change in codon 54 (Pro-Leu) to (Pro-Thr) which were similar and different from those of Asian and North-American, respectively	Targeted sequencing	*pncA* 67%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
4	Huang et al. (2003) [79]	Taiwan	A total of 76 clinical isolates; 59 were PZA susceptible, 17 were found PZA resistant in which 7 isolates exhibited *pncA* mutation	Nucleotide sequence changes were seen in 7 isolates at the *pncA* region. Of these, 4 isolates had nucleotide changes at position 401, 290, 281, 464 and a deletion of nucleotides 352 to 358, a G insert at nucleotide 397, and a Phe 94 → Ser mutation were found	Targeted sequencing	*pncA* 41%
5	Miyagi et al. (2004) [80]	Japan	A total of 26 PZA resistant isolates; 20 had *pncA* mutation, 5 isolates were positive for PZase assay and one isolate was PZase negative and also not found any mutation on *pncA* gene	A total of 15 different mutations were identified and resulted in alteration of the primary amino acid sequence of PZase or frameshifts mutations leading to nonsense polypeptide synthesis Of these, 10 new mutations were not found previously One PZase negative isolate was confirmed with no mutation on *pncA* Five PZase positive isolates also confirmed with no mutation on *pncA*	Targeted sequencing	*pncA* 77%
6	Rodrigues et al. (2005) [81]	Brazil	A total of 59 clinical isolates; 40 were resistant to PZA in which 29 isolates exhibited *pncA* mutation and 11 isolates not had mutation in the *pncA* gene	18 isolates out of 29 had single nucleotide substitutions and 2 of these isolates had single nucleotide substitutions in stop codon at positions 297 and 309 resulting in amino acid replacements 4 isolates had nucleotide deletion and 7 had insertion Of these, 12 new mutations were not described previously	Targeted sequencing	*pncA* 72.5%
7	Louw et al. (2006) [82]	South Africa	A total of 127 clinical isolates; 68 isolates were PZA resistant in which 63 isolates exhibited *pncA* mutation and 5 of them did not have mutation in *pncA* gene	Mutations were found to be diverse and scattered throughout the *pncA* gene 28 different mutations were found that include non-synonymous mutations, insertions and deletions	Targeted sequencing	*pncA* 92%
8	Pandey et al. (2009) [83]	New Zealand	A total of 26 isolates; nine were MDR and 17 were PZA mono-resistant. Of these 11 isolates exhibited mutations in the *pncA* gene	Mutations were detected throughout the length of the *pncA* gene and clustering of mutations were found at amino acidSer18 → His57 and His82 → Val128 regions	Targeted sequencing	*pncA* 42%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
9	Muthaiah et al. (2010) [28]	India	A total of 50 isolates from treatment failure cases; 39 were PZA resistant and 11 were PZA susceptible. Of these resistant strains 33 isolates had *pncA* mutation	The majority of mutations were found in three regions of *pncA:* 3–17(15%), 61–85(30.7%) and 132–142(33.8%), and other mutations were observed at K96 and H51 (7.6%), A171 (2.5%) Another mutation identified at amino acid 26 Ala → Gly (5.1%) of *pncA* which was not reported previously	Targeted sequencing	*pncA* 78%
10	Jonmalung et al. (2010) [84]	Thailand	A total of 150 isolates; 100 were MDR and 50 were susceptible. Among 150 isolates, 49 had *pncA* mutation, of which 39 were PZA resistant and 10 were PZA susceptible	Mutations were found scattered along the coding and promoter regions with high diversity Most frequent mutation (16%) being His71Asp 17 isolates showed either Ile31Ser or Ile31Thr mutations 22 mutation types were detected and showed a correlation with PZA resistance. Of these, 14 nucleotide substitutions and 2 putative promoter region mutations were previously reported Six novel mutation types were identified which are not previously discussed, of these 3 nucleotide substitutions (Leu27Pro, Gly122Ser and Thr174Ile), 2 nucleotide insertions (G insertion between nucleotide 411 and 412 and GG insertion between nucleotide 520 and 521), and 1 nonsense mutation at Glu127 Mutations at Ile31Ser and Ile31Thr were found in both PZA-susceptible and PZA-resistant isolates	Targeted sequencing	*pncA* 75%
11	Chiu et al. (2011) [85]	Taiwan	A total of 66 MDR isolates; 36 were PZA resistant and 30 were PZA susceptible Among 36 PZA-resistant isolates, 29 had *pncA* mutation	39 different mutations were scattered along the whole *pncA* gene; out of 39 mutations, 37 were point mutations, 1 deletion and 1 insertion Also was found one mutation at nucleotide 98Asp(33) → Ala in *pncA* gene of a PZA-susceptible isolate	Targeted sequencing	*pncA* 80.6%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
12	Campbell et al. (2011) [86]	*Georgia, US*	A total of 192 isolate; 65 were PZA resistant and 127 were PZA susceptible. Out of 65 PZA resistant strains, 55 had *pncA* mutation. Out of 127 PZA-susceptible isolates 15 had mutations in *pncA* gene	This study found 79 unique alterations distributed throughout the *pncA* gene and its regulatory region Of these, majority of mutations were 72.4% nSNPs which include 10 isolates that had insertions, 9 had deletions and 6 had nonsense, 3 had synonymous, and 4 had putative regulatory mutations The most frequent mutation was identified at amino acid Thr47Ala in 5 isolates and a synonymous mutation at ser65 in 3 isolates	Targeted sequencing	*pncA* 84.6%
13	Alexander et al. (2012) [38]	Canada	A total of 141 isolates; 77 were PZA susceptible and 64 were PZA resistant, of these 53 had *pncA* mutation and 11 had no mutation in the *pncA* gene	53 isolates exhibited *pncA* mutation, including insertions, deletions, non-synonymous changes, and promoter mutations were observed Out of 11 isolates, a strain had synonymous mutations at Arg212 and Leu320 in the *rpsA* region In addition, identified *rpsA* mutation A364G (Lys-122Glu), was present in one PZA-susceptible isolate This study observed that these *rpsA* mutations have no impact on PZA resistance	Targeted sequencing	*pncA* 82%
14	Bhuju et al. (2013) [87]	Brazil	A total of 97 isolates; 62 were PZA susceptible and 35 were PZA resistant. Of these 16 isolates had *pncA* mutation	Mutations were found throughout the length of the *pncA* gene 14 isolates had point mutations which caused amino acid substitutions near the active site or metal binding site (active site: V7; metal site: D49, C72, T76) Mutation was not found in putative promoter region of the *pncA*	Targeted sequencing	*pncA* 45.7%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
15	Jnawali et al. (2013) [88]	Korea	A total of 30 pan susceptible, 100 XDR and 62 MDR strains were used in this study. Of these, they found 31PZA-susceptible isolates. Among the PZA resistant isolates 107 were detected *pncA* mutation and 22 isolates did not show mutation in the *pncA* gene	This study found that majority of mutations were clustered in the regions between amino acid 3–17, 132–142 and 155–172 and also found 70 different point mutations Most frequent *pncA* mutation was observed in Leu159Arg with Thr135Pro mutation They identified 36 new mutations which were not previously described	Targeted sequencing	*pncA* 87.7%
16	Akhmetova et al. (2014) [23]	Kazakhastan	A total of 71 isolates; 41 had PZA resistant and 36 were PZA susceptible. Among the resistance, 36 had *pncA* mutation and 5 isolates did not show mutation in the *pncA* gene whereas *rpsA* mutations were found in 5 of 3 isolates	This study found 22 different nucleotide substitutions in 22 isolates 3 isolates had Gln10Pro mutations and 3 isolates had Ala171Glu mutations 5 PZA-resistant isolates lacked *pncA* mutations, of these 3 isolates had synonymous mutation at codon 212 in the *rpsA* gene Among the PZA-sensitive isolates, 2 strains had Gly17Cys and Val93Met mutations in the *pncA* gene	Targeted sequencing	*pncA* 87.8% *rpsA* 7.3%
17	Tan et al. (2014) [22]	Southern China	A total of 161 clinical isolates; 109 isolates were PZA susceptible and 52 were PZA resistant. Of these, 44 isolates showed *pncA* mutation remaining 8 isolates did not show mutation in *pncA* gene. Whereas *rpsA* mutations were found in 3 of 8 isolates	This study observed that mutations were dispersed along the *pncA* gene Among the 44 mutant isolates, 33 harbored single point mutation, 8 had deletions and 3 exhibited double point mutations Clustering of mutations of G132-T142 and P69-L85 were observed in 13 of 44 mutants 1 strain had a point mutation in the putative regulatory region (A-11G) of the *pncA* gene 10 new types of *pncA* mutations were found in 13 strains which were not previously described *rpsA* mutations (PZAR-*pncA* WT) were found in 3 isolates which had single point mutation R474L, R474W and E433D all clustered in the C- terminus of RpsA protein One PZA-susceptible isolate also had a Q162R single mutation in the *rpsA* gene	Polycistranic Reverse transcription method	*pncA* 84.6% *rpsA* 5.7%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
18	Gu et al. (2015) [41]	China	A total of 142 isolates; 82 were PZA resistant and 60 were PZA susceptible. Among the resistant isolates, 65 isolates exhibited *pncA* mutation and 26 isolates had *rpsA* mutations	Majority mutations were found non-synonymous single point mutations (84.8%) in 56 isolates and 1 isolate had double point mutation but one of them was synonymous mutation (Glu107Glu, Glu111Lys) Five isolates that had deletions and five that had insertions were also detected Among 60 PZA-susceptible isolates, 6 harbored single point mutations in *pncA* genes, but 1 of them was synonymous mutation 26 of 82 PZA resistant isolates had 24-point mutations and 2 insertions in *rpsA* gene; all were dispersed along the whole gene Among the 60 PZA susceptible isolates, 6 isolates had *rpsA* mutations, and all those were single point mutations A total of 16 strains had mutations both within *pncA* and *rpsA*, and all of those belonged to PZA-resistant strains	Polycistranic Reverse transcription method	*pncA*79.3% *rpsA* 31.7%
19	Maslov et al. (2015) [89]	Russia	A total of 64 isolates; 41 had resistance to any one anti-TB drugs and 23 were drug susceptible. Among the resistants, 28 isolates had *pncA* mutations	This study found 24 unique variants of *pncA* mutations harboring non-synonymous mutations and no synonymous mutations Of which, 3 strains had A188C mutation and 2 strains had A-11G- mutation in promoter region Among the PZA susceptible isolates, 5 had mutations in *pncA* and negative PZase activity This study also found a silent mutation in *rpsA* gene (A636C) but not involved in PZA resistance. Another mutation in codon 432 (T1295C, leading to the amino acid change M432T) of *rpsA* which also is not involved in PZA-resistance Another mutation A479C was found in *rpsA* gene PZA-susceptible isolate The mutation in *panD* gene G415T was found in a drug-susceptible strain This study suggested that mutations in *rpsA* and *panD* are not associated with PZA resistance	Targeted sequencing	*pncA* 80%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
20	Wabale et al. (2016) [30]	India	A total of 100 isolates; 36 were PZA resistant, of which 34 isolates exhibited *pncA* mutation	This study found that 37 variety of mutations were dispersed along the *pncA* gene Some degree of clustering of mutations including both missense and nonsense mutations were found in the 5 strains; 22 strains had amino acid substitution, 10 had nonsense and 2 had *pncA* promoter region	Targeted sequencing	*pncA* 94.4%
21	Reena et al. (2016) [90]	India	A total of 16 isolates; 6 were PZA resistant and 10 were PZA susceptible. Among 6 resistants, 3 had *pncA* mutation	Mutations showed nucleotide substitutions and insertions 80% of the isolates showed deletion of nucleotide at position 67 of the *pncA* gene It was observed that mutations were not dispersed throughout the *pncA* gene	Targeted sequencing	*pncA* 50%
22	Huy et al. (2017) [91]	Vietnam	A total of 260 isolates; 123 were multidrug resistance (MDR). 99 of 260 isolates exhibited *pncA* mutations of which 89 isolates were MDR	There are 71 different types *pncA* mutations found in the *pncA* coding region (32.3%) or its promoter (3.9%) Non-synonymous mutations were detected in the *pncA* coding region and were dispersed throughout the gene They detected most common *pncA* promoter mutation of a nucleotide substitution at position − 11, − 12, − 13 and a 12-nucleotide deletion (from − 18 to − 7) were also detected Single nucleotide substitutions were found at codons 103, 108, 119, 164 or 181 which resulted in premature stop codons 30 of 91 mutants carried *pncA* mutations in the three regions described as mutation hot spots (i.e., 3–17, 61–85 and 132–142) 16 new mutations were found at codons 2 (Arg-Trp), 47 (Thr-Ile), 58 (Phe-Val), 82 (His-Gln), 142 (Thr-Arg) and 164 (Ser-Stop); these are not previously described	Targeted sequencing	*pncA* 72%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
23	Liu et al. (2017) [39]	Hangzhou, China	A total of 291isolates; 161 were MDR strains, of which 145 exhibited *pncA* mutation and 16 isolates did not show mutation in *pncA* gene. Whereas 10 of 16 isolates exhibited *rpsA* mutation and 3 isolates had *panD* mutation	A total of 155 isolates had non-synonymous or InDel mutations in *pncA* and its promoter region 34 isolates had InDels mutation (22%), 3 had stop codon mutations (1.9%), 8 had mutations in promoter (5.2%) regions and 110 had single non-synonymous mutations (71%) located throughout the entire *pncA* gene 12 isolates had 9 different kinds of mutations that were observed in the *rpsA* gene 4 isolates exhibited non-synonymous mutations in *panD*, of which 1 had both *pncA* and *panD* insertion mutations and remaining 3 had only exhibited a C400T mutation in panD *PncA* D12G, L27P and V155A mutations found in both PZA-susceptible and PZA-resistant isolates	Targeted sequencing	*pncA* 90% *rpsA* 7.5% *panD* 2.5%
24	Rahman et al. (2017) [92]	Bangladesh	A total of 169MDR TB isolates; 76 were PZA resistant. Among 169 MDR 87 strains showed *pncA* mutation and 65 of them did not show *pncA* mutation	There were 64 different mutations found in both putative promoter region and the *pncA* gene 68 isolates were harbored single mutation, 9 had double mutations and 10 had silent mutation (Ser65Ser) Two strains had mutation at A(-11)G and A(-15)C in the promoter region Three isolates had a long segment deletion (del Cd37-187), 7 isolates had two bases insertion (445 ins GC) and a silent mutation (TCC-TCT, Ser65Ser) was found in 13 isolates. This study detected 27 new mutations that which are not reported previously	Targeted sequencing	*pncA* 57.2%
25	Daum et al. (2018) [75]	Ukraine	A total of 91 isolates; 67 isolates exhibited *pncA* gene mutation while 24 were determined to be 'wild-type' by WGS	45 different mutations were found in the *pncA* coding and upstream promoter region They found that most of the mutations were single mutations, and also found amino acid substitution, stop-translation, deletion, insertion, nucleotide frameshifts at various positions This study detected 11 new mutations in the *pncA* which were not previously reported	Whole genome sequencing	*pncA* 74%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
26	Khan et al. (2019) [93]	Pakistan	A total of 69 PZA-resistant isolates; 52 were MDR and 6 were XDR strains. Of these 51 strains had *pncA* mutation. A total of 26 PZA-susceptible isolates were analyzed	This study found 36 different mutations in coding region of *pncA* The most common mutations were detected at positions 287 and 423 (Lys96Thr & Ser179Gly) and common synonymous mutation at position 195C > T was observed in both PZA-resistant and PZA-susceptible isolates This study found 15 new mutations in *pncA* gene including 194_203delCCTCGTCGTG and 317_318delTC detected that were not reported in TBDRM and GMTV databases	Targeted sequencing	*pncA* 74%
27	Tam et al. (2019) [71]	China	A total of 326 *Mycobacterium tuberculosis* complex isolates; 266 were pan susceptible and 27 isolates were PZA resistant of which all the 27 isolates had mutation in the *pncA* gene	Mutations were scattered along the entire *pncA* gene without any hotspot regions including Phe13Ser in 1 isolate, Cys14Arg in 1 isolate, Ser66Pro in 3 strains, Trp68Arg in 1 strain, Ile90Ser in 2 strains, Val125Aspin 1 strain, Gly162Asp in 6 strains and Leu182Ser in 1 strain This study found 6 new *pncA* mutations, these included 5 with mutations (Del383T, Del380-390, A-Insertion at 127, A-Insertion at 407 and G-Insertion at 508) in *pncA* structural gene, and 1 with mutation (T-12C) at *pncA* promoter region	Targeted sequencing	*pncA* 100%
28	Xia et al. (2020) [94]	China	A total of 324 isolates; 132 were PZA resistant of which 97 isolates exhibited *pncA* mutation and 16 isolates were PZA susceptible	This study found that mutations were scattered throughout the entire *pncA* gene and that there were no hot spot regions 61 different point mutations were observed causing amino acid change and 11 frameshift changes Four-point mutations were Met1Thr, Asp12Asn, His137Arg, Thr47Ala detected both in PZA-resistant and PZA-susceptible isolates 15 non-synonymous mutations, 1 frame shift mutation was detected in PZA-susceptible isolates This study found 21 new mutations which were not reported earlier	Targeted sequencing	*pncA* 73.4%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
29	Hameed et al. (2020) [33]	China	A total of 448 isolates; 269 were PZA resistant and 179 were PZA susceptible. Among 269 isolates, 227 isolates exhibited *pncA* mutation, 7 had *rpsA* mutation, 4 had *panD* mutation, 3 had *Rv2783c* mutation and 2 had *Rv2044c* mutation	This study found mutation in the upstream flanking region (UFR) of *pncA* (Rv2044c-*pncA*-Rv2042c); among 227 PZA-resistant isolates, 134 strains carried single non-synonymous mutation, 54 strains had exhibited insertions/deletions (indels) in *pncA* coding region, 13 strains had stop codon mutations, 1 strain had multiple mutations, 2 strain had synonymous mutation and 12 strains had mutations both in *pncA* and its UFR In the *rpsA* gene, one non- synonymous mutations Thr29Met and 2 new mutations, Gln162Arg, Ala412Val, were detected They observed synonymous mutation Arg212Arg in the *rpsA* gene of both PZA-resistant and PZA-susceptible isolates In C-terminus of *panD* gene, mutations Leu-132Pro (Novel) and Pro134Ser were detected in PZA-resistant strains A new non-synonymous mutation Ser149Pro was detected in Rv2783c of PZA-resistant strains The non-synonymous mutations Tyr70His, Ile71Asn, Trp80Gly and Trp80Leu, Ala82Asp were found in *Rv2044* (the instant upstream gene of *pncA*) Over all this study identified 57 (23.45%) new non-synonymous mutations/indels	Targeted sequencing	*pncA* 83.64% *rpsA*-2.6% *panD*-1.48% *Rv2783c*-1.1% *Rv2044c*-0.74%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
30	Shi et al. (2020) [44]	Henan Province, China	A total of 152 MDRTB isolates; 105 were PZA resistant and 47 were PZA susceptible. Among 105 isolates, 102 exhibited *pncA* mutation, 76 had *rpsA* mutation and 2 had *panD* mutation. 35 of 47 PZA-susceptible isolates had mutation in the *rpsA* gene	This study identified a total of 100 different *pncA* mutation patterns, including 80-point mutations and 20 insertions/deletions, throughout the entire length of the *pncA* gene and its promoter region and also in open reading frames 89 strains harbored a single *pncA* mutation, either a single base or a synonymous mutation and 13 strains had multiple mutation 32 new *pncA* mutation patterns were detected including 16 substitution mutations and16 frame shift mutations They found *rpsA* mutations that were scattered and located in the180 amino acids of the N-terminus and C-terminus region Two strains harbored *rpsA* codon357 and codon532 mutations without *pncA* mutation and a non-specific synonymous mutation CGA–CGC in *rpsA* codon 212 was identified in both PZA-resistant and PZA-susceptible isolates Two stains harbored Val56Ala and Glu130Gly mutations in the *panD* gene	Targeted sequencing	*pncA*-97% *RpsA*-72% *panD*-0.9%
31	Naluyange et al. (2020) [95]	Uganda	A total of 159 isolates; 12 were PZA susceptible and 25 were PZA resistant, of the resistant 6 strains had *pncA* mutation	This study found most common mutation of V180F in four isolates and V180F and T142R double mutation identified in one strain, another mutation Lys96Arg in one strain and Arg154Gly in one strain No mutation was observed in PZA-susceptible strains	Targeted sequencing	*pncA* 24%
32	Li et al. (2021) [96]	Chongqing, China	A total of 465 isolates; 346 exhibited *pncA* mutation. Of the 465, 294 isolates were PZA resistant	This study found 124 types of mutations which include one synonymous mutation and 123 non-synonymous mutations Three mutations were observed at positions – 12 and – 11 upstream of *pncA* and the remaining 121 mutations were distributed within the *pncA* open reading frame There were found 30 new mutations which were not reported previously Most of the mutations were scattered between nucleotide position 21–521	Targeted sequencing	*pncA* 97.4%

Table 1 (continued)

S.No	Author	Region	Isolates selection	Nature of resistant mutations	Method of mutation identified	Frequency of mutations
33	Che et al. (2021) [97]	Ningbo, China	A total of 110 MDRTB isolates; 45 were PZA susceptible and 65 were PZA resistant. Of these, 54 showed mutations in the *pncA* gene	There were 40 different mutant types observed in the *pncA* gene, including 91.1% of single nucleotide substitutions and 8.9% of frame shift mutation	Targeted sequencing	*pncA* 83%
34	Modlin et al. (2021) [58]	Sweden	A total of 19 phenotypically PZA-mono-resistant isolates were collected. Of these, 10 isolates showed *pncA* mutation and remaining 9 isolates were WT *pncA* with PZAR. Among these two had *panD*, two had *rpsA* and one isolate was detected *M. canettii*. 11/19 clinical isolates showed *clpC1* mutation	This study was reported the first *clpC1* mutations in PZAR clinical isolates with WT*pncA* and a *panD* mutation (*panD*Asp133Ala) in its new C-terminal recognition tag novel to clinical isolates The first clpC1 mutation is a deletion in the clpC1 promoter (*clpC1p*), 138 bp upstream of its start codon (*clpC1p* – C138) and the second *clpC1* mutant was a missense mutation in the *ClpC1* N terminus, Val63Ala ($clpC1_{\mathrm{Val63Ala}}$) A large deletion was detected in one isolate in the beginning of *pncA* gene using WGS	Targeted sequencing and WGS	NA[a]

Abbreviations: PZA- pyrazinamide, *PZase*- pyrazinamidase, *SSCP*- single-strand conformation polymorphism, *MIC*- minimum inhibitory concentration, *nSNPs*- non-synonymous single nucleotide polymorphisms, *InDels*- insertion/deletion polymorphisms, *MDR TB*—multi-drug resistant tuberculosis, *GMTV*- genome-wide *Mycobacterium tuberculosis* variation, *NA* not applicable

[a]NA- The total number of isolates screened to select 19 PZAR isolates is not given in the paper

is involved in the rescue of stalled ribosomes during mRNA decoding. Trans-translation is related to the restoration of stress survival, virulence and nutritional deficiencies. When POA binds to RpsA, it disrupts the interaction between RpsA and tmRNA. The blockade of the trans-translation pathway due to POA leads to defective rescue of stalled ribosome which results in the depletion of available ribosomes, in addition to which it causes the accumulation of toxic or deleterious proteins, affecting persister survival under stress conditions [35].

rpsA Gene Mutation

In 2011, the C-terminal of the RpsA protein was stated as a novel target of POA by Shi et al. However, one study concluded that RpsA is not a PZA target [36]. It is believed that the *rpsA* mutations play an accessory role in PZA resistance without *pncA* mutation [37]. Whereas Alexander et al. reported that PZA-resistant isolates had no *rpsA* mutations however a single non-synonymous *rpsA* mutation was seen (Lys122Glu) and one synonymous mutation (Arg212) in two PZA sensitive strain and that *rpsA* gene along with flanking region is inconvenient for amplification and sequencing owing to the gene length. Therefore, the analysis of the *rpsA* gene had no role in the detection of PZA resistance [38]. Subsequently, a study found three novel non-synonymous *rpsA* mutations in PZA-resistant isolates that supported the *rpsA* mutation as a new mechanism, which included amino acid substitutions like Arg474Trp, Arg474Leu and Glu433Asp [22]. Liu et al. reported non-synonymous mutations in *rpsA* gene of 12 PZA-resistant isolates and suggested that further analysis is needed to find the impact on detection of PZA resistance [39]. Another study found a synonymous mutation Arg212Arg of *rpsA* in three PZA-resistant strains, and these strains were clonal belonging to a genetic cluster profile. This leads to the assumption of the presence of other mechanisms of PZA resistance other than *rpsA* gene mutation [23]. A study by Yang et al. in 2015 described that PZA resistance has some association with the crystal structures of C-terminal domain of RpsA, POA, and two RpsA variants [40]. The structural analysis further disclosed that hydrogen bonds and hydrophobic interactions were involved in the binding of POA to RpsA, it and was mediated by conserved residues (Lys303, Phe307, Phe310, and Arg357). Mutation and sequence variation at the C-terminus of RpsA can lead to loss of POA binding. Gu et al. has described further mutations in clinical strains, where maximum mutations occurred at C-terminal region [41]. Furthermore, Khan et al. established the location of majority of *rpsA* mutations in nucleotides 973–1051 coding region, specifically at 1024–1030 [42]. In a recent study, 2.6% PZA-resistant strains carried non-synonymous mutation Thr29Met and two newly identified mutations Gln162Arg and Ala412Val in *rpsA* [33]. However, another study hypothesized that *rpsA* Δ438A mutation is not involved in PZA resistance, which raises doubts against consideration of *rpsA* as a target of PZA [36]. But Shi et al. produced evidence that were in contrast with the above findings, they showed that the mutations *rpsA*Δ438A and D123A were involved in PZA resistance and provided further proof that the *rpsA* can be considered as a real target for PZA. The study also reported that 72% of *rpsA* mutations are dispersed, where maximum mutations were located in the 180 amino acids of the N-terminus and C-terminus of RpsA [43]. A high frequency of synonymous mutations at codon 212 was observed in PZA resistance and PZA sensitive strains, and no nonsense or deletion/insertion mutations were found in *rpsA* of PZA-sensitive strains. This shows that additional research should be carried out to investigate the effects of the new mutations on PZA resistance [44]. However, another study denied the capability of POA to compete with tmRNA. *rpsA* wild type (WT), *rpsA* ΔA438 and *rpsA* ΔA438 variants (truncated at the carboxy terminal end) were analyzed, and it was shown that there is no competition between POA and tmRNA, since no displacement of tmRNA from *rpsA* or any of its variants was observed in the presence of excess POA [45]. Although few *rpsA* mutations were found to be associated with PZA resistance, most of these *rpsA* mutations were reported in PZA-sensitive strains [46]. Strain DHMH 444 with ΔAla438 mutation became the only previously reported mutant in the clinical strain. The strain was labeled as "less susceptible to PZA, with low-level resistance (MIC, 200 to 300 mg/mL)" [47]. Limited availability of data causes hindrance in establishing the association between *rpsA* mutations and PZA susceptibility. Therefore, the involvement of *rpsA* in PZA resistance is inconclusive and debatable at present.

PanD Action

panD is a gene which encodes aspartate decarboxylase (139 amino acids, 15KD protein), aids in the formation of β-alanine from L-aspartate, additionally involved in the synthesis of pantothenate (vitamin B5) and coenzyme A (CoA), an enzyme that is crucial for many of the energy metabolism where carbohydrates, fats and proteins are used as energy sources in *M. tuberculosis* [48]. Zhang et al. identified that PZA inhibits PanD in *M. tuberculosis* leading to inhibition of formation of those essential cofactors [49]. Further findings revealed that when PanD was overexpressed, it caused significant resistance towards POA and PZA. The PanD enzyme activity was inhibited by POA in a concentration-dependent manner at curatively significant concentrations, but was not affected by the prodrug PZA. Further it was observed that the anti-tuberculosis activity of POA was counteracted by the presence of alanine or pantothenate in the 7H11 growth medium in PZA-susceptible strains. This raises intriguing questions on whether this observation has significant repercussions in the potential food-derived antagonization of POA through the diet of the patient in vivo. These findings indicate that PanD is a new target for POA/PZA [50]. Dillon et al. disputed that mutation in *panD* does not confer PZA resistance, since the pantothenate-auxotrophic strain was cultured employing a sub-antagonistic concentration of pantetheine as well as pantothenate [51]. However, a subsequent study has shown that PZA/POA has at least two discrete mechanisms of action and resistance. First comprises the resistance caused by *panD* mutations that did not affect CoA synthesis. The second mechanism involves phthiocerol-dimycocerosate (PDIM) synthesis, and its resistance is caused by loss-of-function mutations in PDIM synthesizing genes *mas* (mycocerosic acid synthase) and *ppsA*-E (phenolpthiocerol synthesis type-I polyketide synthases). These genes prevent [52] toxicity and facilitate growth of *M. tuberculosis* in vitro even when the drug and structural analogues were present. The excellent in vivo efficacy of PZA is due to the synthesis of a virulence factor, but still a clear picture cannot be arrived at due to the lack of clear molecular resistance and epidemiology of PZA [52].

panD Gene Mutation

In some PZA-resistant strains, mutation was evident in *panD* but not in *pncA* or *rpsA* [49]. The natural resistance of *M. canettii* to PZA can be attributed to a non-synonymous Met117Thr change and silent mutation C39G in *panD*. In MDR-TB clinical strains, amino acid substitution of Pro134Ser in *panD* was responsible for PZA resistance. Shi et al. using 30 POA mutants, concurred that *panD* mutations localized to the C-terminus of the PanD protein, particularly the Met117 residue conferred PZA resistance and that this is the most common site altered in POA-resistant mutants, leading to inhibition of panD enzymatic activity [50]. Several studies have revealed *panD* to be a PZA target, and various reports investigated *panD* mutations in *M. tuberculosis* clinical strains. In a study by Werngren et al. WGS revealed extensive analysis of 26 M. *tuberculosis* isolates selected as PZA resistance with wild-type *pncA* gene. The two strains showed non-synonymous Ile49Val and Ile115Thr mutations in *panD* [37]. Shi et al. demonstrated *panD* mutation in two (1.9%) PZA-resistant strains, among which, one had wild-type *pncA* [44]. Hameed et al. detected 1.48% PZA-resistant strains with two mutations, Leu132Pro and Pro134Ser at the C-terminal end of PanD, which could be a possible reason for the alteration of the protein assembly and development of resistance. The mutation at Leu132Pro is considered as a unique mutation in *panD* [33]. The present hypothesis is that resistance to PZA/POA is caused by mutation in the active site loops, among which recombinant protein was purified from two resistant mutants, His21Arg and Met117Ile. The low-binding affinity of POA to Met117Ile, when compared to wild type of *panD*, suggests resistance due to these two mutations depends on a combination of lower affinities for the reduced interaction time of POA in *panD* [53].

Other Genes Action and its PZA Resistance

Newly emerging mutations in genes like *clpC1, mas* and *ppsA-E* were not prominent in clinical isolates. But when analyzed in vitro, some POA/PZA-resistant strains exhibited mutation in *clpC1, mas* and *ppsA-E*; showed impaired growth in vivo, and this potentially clarifies their apparent absence in PZA-resistant clinical isolates. In contrast, in vivo mutations were indistinguishable when compared with wild-type showing POA/PZA-resistant *panD* mutant [54]. Caseinolytic protein C (ClpC), a general stress protein which is found in prokaryotes and eukaryotes, has been related to bacterial pathogensis. ClpC1 (ClpC homolog) in *M. tuberculosis* is involved in protein degradation by forming a complex with protease clp1 and clp2, and it recognizes misfolded proteins [55]. By using comparative sequence analysis, Zhang et al. found out that PZA-resistant mutant S14 was deficient in *pncA, rpsA* and *panD* mutations. This mutant S14 strain had a single unique mutation G296T, which resulted in an amino acid change to Gly99Asp in the new gene *clpC1* (Rv3596c), encoding an ATP-dependent ATPase when compared with parent strain *M. tuberculosis* H37Rv [56]. Concurrently, it was found that POA-resistant strains harbored mutations in two different regions of the ClpC1 protein [57]. In 2021, Modlin et al. found *clpC1* mutations in the promoter (clpC1p -138) and N-terminal (*clpC1*Val63Ala) regions of PZA resistant clinical isolates. But the precise role of *clpC1* in the direct mechanism involved in the action of PZA/POA and the resistance mechanism has not yet been elucidated [58]. Other genes like *mas* and *ppsA-E* take part in the cell wall lipid phthiocerol dimycocerosate (PDIM) synthesis. PDIM is a virulence factor required for the growth of *M. tuberculosis* in vivo, but not in vitro. Since the PDIM genes *mas* and *ppsA-E* mainly exhibited frameshift mutations dispersed through diverse domains, it led to loss-of their function in vitro [52]. In the subsequent study, it was found that POA/PZA-resistant strains harboring *clpC1, mas* and *ppsA-E* mutation in vitro displayed reduced growth in vivo. This provided a promising elucidation for their absence in PZA-resistant clinical isolates highlighting the difficulty in detecting *clpC1*, and *mas/ppsA-E* mutations among PZA-resistant clinical isolates for better understanding [54].

A study in China characterized the genomic DNA of 109 POA-resistant mutants. These mutants were sequenced for the following genes *pncA*, *rpsA*, *panD* and *clpC*. Significantly, 101 of 109 POA-resistant mutants produced 93% *panD* mutations, and four had *clpC1* mutations (Ser91Gly). New mutations were identified in LprG (*rv1411c*) and *rv0521*, *rv3630*, *rv0010c*, *ppsC* and *cyp128.* The dual function of enzyme Rv2783 was also described as a new target of POA [59]. Substitution mutation in *Rv2783c* (A2104C) was reported in PZA-resistant clinical isolates which confers PZA resistance. In the wild type, they found that all catalytic events of Rv2783 were inhibited by POA, and it was observed that there was competition between POA tmRNA for binding with Rv2783. This was not observed in the mutant. Whether the binding of tmRNA to Rv2783 is essential for *M. tuberculosis* or not is not clear [60]. The author suggested that further research is needed to validate the involvement of identified mutations in PZA resistance.

In summary, mechanism of PZA resistance is mainly presided by *pncA* mutation, because the mutations are scattered and highly diverse. In addition, most of the studies from high TB burden regions showed that the PZA resistance was due to mutations in *rpsA* and *panD*, and these were observed only in MDR strains without *pncA* mutations. The involvement of *rpsA* and *panD* mutations in PZA resistance is still debatable, therefore additional experimental data is necessary to illuminate the role of *rpsA* and *panD* in the development of PZA resistance. Overall, many studies detected new mutations and new mechanism of resistance but still conclusive evidences are needed. Also, the molecular targets and mechanisms have not completely described the activity of PZA, and additional mechanisms that are not explored need to be considered.

WHO Mutation Catalog and Advanced Clinical Trials

The mutation catalogue of WHO is a significant milestone for understanding and detection of mutation. The WHO mutation catalogue used vast sample size (> 38,000) of MTBC isolates sourced from multiple countries, and it assembled WGS and Phenotype data to report the resistance-associated genetic variants for predicting clinically relevant resistance phenotypes [61]. This could provide standardized reference for interpretation of resistance to all first line and second line drug resistance which might be of help in developing methods to identify PZA resistance and aid in designing rapid diagnostic tools that can be easily used in facilities with even the minimal infrastructure.

There are several on-going clinical trials for TB therapy. Pyrazinamide can facilitate to shorten the present chemotherapy regimen when used in combination with all new TB regimes as summarized in Table 2.

Pyrazinamide Susceptibility Testing (PST)

In PZA susceptibility testing, a proportion of both resistant and sensitive strains are detected and evaluated with reference strains using phenotypic assay. Pyrazinamide requires acidic pH for activity, and the growth of *M. tuberculosis* is inhibited if pH of the medium is below 5.5. *M. tuberculosis* is susceptible to acidic pH when compared with other mycobacteria [62]. In vitro PST which involves standard agar-based methods frequently generate uncertain outcomes on account of substandard absorption of test media, the use of inappropriate acidic medium pH, and excessively large inoculum which can diminish the PZA activity generating false resistance [18]. Whereas the use of several rapid phenotypic methods which include radiometric BACTEC 460, flourimetric Mycobacteria Growth Indicator Tube (MGIT) 960, ESP Culture System II and the colorimetric BacT/ALERT 3D system ensued as very useful for fast and consistent susceptibility testing of *M. tuberculosis* isolates [63]. Currently many laboratories that use non-radiometric, fully automated, continuous-monitoring MGIT 960 system for performing liquid DST give consistent result with a remarkable processing time of 15–22 days [30]. Conventional culture-based methods involve many challenges that make it unreliable [64], however, PST method is not routinely performed in many countries.

Pyrazinamidase activity assay allows the identification of PZase enzyme activity and is a better alternative to the conventional procedures. The presence of active PZase enzyme is determined by the hydrolysis of PZA to POA which can be observed by the change in color [65]. Though the assay can misinterpret results as false resistance, some isolates with PZase activity have been shown to be PZA resistant (resistance other than a mutation affecting PZase). Pyrazinamidase assay sensitivity has been reported to vary between 79 and 96% [66]. In a study, 9% of wild-type strains were found to be PZA resistant with *pncA* gene, it was also shown that eight strains were positive for PZase activity and eight strains were negative during PZase assay. This provides the evidence of involvement of other genes which can regulate and control the expression of PZase and other unknown mechanisms of PZA resistance [66].

Predominantly, MGIT system has produced varying test results while evaluating PZase activity to check the presence of *pncA* mutations [67]. To overcome these difficulties, molecular diagnostic assays and the infrastructural containment requirements for culture-based assays can prove to be helpful in getting uncompromised results.

Table 2 Clinical trials for TB therapy

Serial No	Trial ID	Criteria	Number of subjects	Phase	Intervention	Study sponsor	Status	Remarks/findings
1	NCT05047055 https://clinicaltrials.gov/ct2/show/NCT05047055	18–65 years. Sputum smear/CBNAAT test positive for tubercle bacilli	550	NA	Isoniazid Rifampicin Pyrazinamide Ethambutol Moxifloxacin	India	New/not yet recruiting	NA
2	NCT00933790 https://clinicaltrials.gov/ct2/history/NCT00933790	18 years and older. HIV-1/2 infected patients with pulmonary TB and sputum smear positive	331	3	Isoniazid Rifampicin Pyrazinamide Ethambutol	India	Completed	Among HIV-positive patients with pulmonary TB receiving antiretroviral therapy, a daily anti-TB regimen proved superior to a thrice-weekly regimen in terms of efficacy and emergence of Rifampicin resistance
3	NCT00864383 https://clinicaltrials.gov/ct2/show/results/NCT00864383	18 years and older. Newly diagnosed and previously untreated *M. tuberculosis* infection, sputum smears positive with culture-confirmed susceptibility to Rifampin and fluoroquinolones	1548	3	Moxifloxacin Ethambutol Isoniazid Pyrazinamide Rifampicin	India	Completed	This study showed that two moxifloxacin-containing regimens produced a more rapid initial decline in bacterial load, when compared to the control group. However, non-inferiority for these regimens was not shown, which indicates that shortening the treatment to 4 months was not effective in this setting
4	NCT02700347 https://clinicaltrials.gov/ct2/show/results/NCT02700347	21–70 years. healthy volunteers	12	1	Pyrazinamide Allopurinol	Singapore	Completed	This study found that the co-administration of Pyrazinamide with Allopurinol did not enhance the whole blood bactericidal activity and is not a useful strategy for increasing the efficacy of PZA in clinical practice
5	NCT04504851 https://clinicaltrials.gov/ct2/show/results/NCT04504851	18–75 years. Abnormalities on CXR compatible with pulmonary TB, sputum smear positive and positive on testing by Xpert MTB/RIF or Xpert Ultra	154	II	Rosuvastatin Rifampicin Isoniazid Pyrazinamide Ethambutol	Singapore	Ongoing	Not available

Table 2 (continued)

Serial No	Trial ID	Criteria	Number of subjects	Phase	Intervention	Study sponsor	Status	Remarks/findings
6	NCT04694586 https://clinicaltrials.gov/ct2/show/results/NCT04694586	18 years and older. Positive for Mtb culture/positive PCR Mtb-complex	40	2	Isoniazid Rifampicin Pyrazinamide Ethambutol	Sweden	Recruiting	NA
7	NCT01498419 https://clinicaltrials.gov/ct2/show/results/NCT01498419	18–65 years. Patients with pulmonary TB and sputum smear positive before treatment	207	2	Moxifloxacin Pretomid Pyrazinamide Rifafour	South Africa	Completed	This study found that the combination of Moxifloxacin, Pretomanid and Pyrazinamide was safe, well tolerated, and showed superior bactericidal activity in drug-susceptible tuberculosis during 8 weeks of treatment
8	NCT04629378 https://clinicaltrials.gov/ct2/show/results/NCT04629378	18–65 years. Sputum smear positive, previously untreated and rifampicin-susceptible pulmonary TB	22	2	Meropenem injection Amoxicillin Clavulanate Pyrazinamide Bedaquiline Rifafour	South Africa	Recruiting	NA
9	NCT03800381 https://clinicaltrials.gov/ct2/show/results/NCT03800381	3 months–14 years. Active TB with or without HIV coinfection, sputum smear positive	92	NA	Isoniazid Rifampicin Pyrazinamide	Ghana	Recruiting	NA
10	NCT04717908 https://clinicaltrials.gov/ct2/show/results/NCT04717908	18–70 years. Sputum smear positive and GeneXpert positive for Rifampicin and Isoniazid	200	NA	Bedaquiline Pyrazinamide Linezolid Cycloserine Clofazimine	China	Recruiting	NA

Table 2 (continued)

Serial No	Trial ID	Criteria	Number of subjects	Phase	Intervention	Study sponsor	Status	Remarks/findings
11	NCT03474198 https://clinicaltrials.gov/ct2/show/results/NCT03474198	18–65 years. Pulmonary TB on chest X-ray and sputum GeneXpert test positive	900	2 and 3	Rifampicin Isoniazid Pyrazinamide Ethambutol Linezolid Clofazimine Rifapentine Levofloxacin Bedaquiline	Indonesia	Recruiting	NA
12	NCT04260477 https://clinicaltrials.gov/ct2/show/results/NCT04260477	No age limit (adults as well as children). Newly diagnosed, sputum smear positive recurrent pulmonary TB	370	3	Isoniazid Rifampicin Pyrazinamide Ethambutol	Belgium	Recruiting	NA
13	NCT03057756 https://clinicaltrials.gov/ct2/show/results/NCT03057756	15 years and older. Never been treated with second line anti-TB drugs for more than 1 month	23	NA	Kanamycin Moxifloxacin Prothionamid Isoniazid Clofazimine Ethambutol Pyrazinamide	Gabon	Recruiting	NA
14	NCT02754765 https://clinicaltrials.gov/ct2/show/results/NCT02754765	15 years and older. Pulmonary TB with resistance to rifampicin and susceptible to fluoroquinolones	750	3	Bedaquiline Delamanid Clofazimine Levofloxacin Moxifloxacin Linezolid Pyrazinamide	France	Recruiting	NA

Table 2 (continued)

Serial No	Trial ID	Criteria	Number of subjects	Phase	Intervention	Study sponsor	Status	Remarks/findings
15	NCT03678688 https://clinicaltrials.gov/ct2/show/results/NCT03678688	18–64 years. Newly diagnosed, drug-susceptible pulmonary TB and sputum smear positive	118	1 and 2	Isoniazid Rifampicin Pyrazinamide Ethambutol Bedaquiline Delamanid	United States	Recruiting	NA

Whitfield et al. does not find any PZA resistance at critical concentration of 100 µg/mL during analysis in MGIT 960 system. It was observed that maximum number of *pncA* polymorphisms were associated with susceptible isolates which were identified in the study and had PZA MIC below the critical concentration (50 and 100 µg/mL) [2]. Pyrazinamidase activity of 136 clinical isolates of *M. tuberculosis* presented 67 resistant strains in MGIT 960, but only 52 were sensitive by both the methods. Conflicting results were observed in 17 samples. Concordance of 88.9% between MGIT 960 and classical Wayne's method was found and 96% PZase positive strains susceptible to PZA were elucidated by Zhou et al., whereas Aono et al. reported 100% PZase positive strains to be susceptible to PZA [65, 68].

A line probe assay is a molecular detection (Genoscholar PZA-TB II) of PZA resistance, which was created by the Nipro (Osaka, Japan). It contains a PZA strip which includes 48 probes enfolding the whole coding region and 18 nucleotides upstream of *pncA*, and it has given high sensitivity (93.2%) and specificity (91.2%) in detection of PZA resistant TB [69]. However, PZA line-probe assay is currently unavailable in many countries.

Molecular detection of PZA resistance in *M. tuberculosis* on the basis of *pncA* mutations are considered more precise when compared to conventional methods [6]. It is more rapid and fast with respect to the traditional mycobacterial susceptibility testing methods which solely depend on the growth of *M. tuberculosis* with disadvantages that create obstacles in testing PZA activity [22]. The whole process of polymerase chain reaction (PCR) and DNA sequencing is around 43 h, but in the conventional method the cultures take around 26 days to grow [70].

PCR mediated Sanger DNA sequencing molecular method is considered as the gold standard detection of PZA resistance. Pyrazinamidase action is presumed by evaluating changes in *pncA* sequence. Pyrazinamidase is considered active when amino acid substitution is not detected along the coding sequence and with no base substitution detected in the promoter sequence, or else it is considered inactive or absent [66]. This was supported by Tam et al. [71]. The relationship between resistance in PZA, action of PZase and the presence of mutations in *pncA* was analyzed. Out of 50 clinical isolates, 39 contained *pncA* mutations, whereas in 11 PZA-resistant strains no *pncA* mutations were detected which can be due to false resistant mutations; these findings were revealed by DNA sequence analysis [28].

Streicher et al. in 2014 showed that 87–95% of PZA phenotypic resistant isolates sheltered non-synonymous mutations in the *pncA* gene [70]. Moreover, Aono et al. reported certain unique findings, they showed that during PCR amplification three *M. tuberculosis* strains did not yield *pncA* PCR amplicons; this was due to the complete deletion of *pncA*, whereas the MICs exceeded 1600 mg/mL. The following strains had deletion at 4475-bp from Rv2041 to Rv2046; a 1565-bp deletion from Rv2042 to Rv2044; and a 6258-bp deletion from Rv2037 to 351 Rv2045, respectively. This proposed the deletion of entire *pncA* leading to development of PZA-resistance. The following sequencing findings stipulate the essentiality of a proper comprehension of PZA resistance with respect to the relation among PZA resistance, PZase action and *pncA* mutations [68].

Compared with MGIT 960 PZA DST, *pncA* sequencing is significantly more sensitive and specific. MGIT 960 PZA DST loses its resistance to PZA at a low-level due to non-synonymous mutations in *pncA*. Though sequencing of *pncA* also has some flaw which can give false resistance to *pncA* mutations, there is significantly no alteration in PZase action and false susceptibility linked to resistance mutations outside *pncA*, but when compared to other molecular techniques, it is less burdensome and more affordable [6].

Recent study reported that the sensitivity rates of PZase assay was 63.1% and that of sequencing methods was 84.2% when compared with the proportion method, while the specificity rates were 80 and 90% by these two methods [72].

Lately, next generation sequencing (NGS) is replacing Sanger sequencing; since it can sequence large amount of DNA fragments concurrently, whereas Sanger sequencing can sequence only small pieces of DNA around 300–1000 base pairs [73]. Next generation sequencing by means of whole-genome sequencing (WGS) and targeted gene sequencing (TGS) enables sequencing of multiple samples with extensive nucleotide coverage depth, mixed strain mutations detection and PZA hetero resistance, that is, drug resistant and drug-sensitive mutations can be also analyzed. Consequently, NGS can facilitate well-timed detection of PZA resistance, as the turnaround time for *pncA* TGS is reported to be roughly 24 h, but only limited studies have focused on genetic diversity of mutations involved in *pncA* gene of clinical strains.

Through Ion Torrent next-generation sequencing, PZA resistance in *M. tuberculosis* isolates was detected and was compared with phenotypic and genotypic (Sanger method) results. Concurrence between NGS and MGIT results was 93%, but the values among Wayne test, *pncA* gene Sanger sequencing and NGS were 82%, 83% and 89%, respectively. Therefore, the author proposed that NGS established majority of *pncA* mutations when compared to Sanger sequencing, but also identified several new and mixed strain mutations that helped to identify conflicting outcomes from traditional methodologies [74].

In a Ukrainian study, genetic analysis was performed using NGS to describe resistant mutations in the *pncA* gene. Mutations were sensed in 67 of 91 clinical strains, including 45 mutation types in the *pncA* coding and upstream promoter region; 11 novel *pncA* mutations were recorded. Three of them were established to be PZA-resistant, seven isolates consisted mixed base mutations and four harbored double mutations. These studies favored the use of NGS for *pncA* gene characterization and proposed its contribution in PZA therapy, particularly in MDR- and XDR-TB patients [75]. However, corresponding phenotypic DST data is needed for validation and confirmation of PZA resistance in addition to NGS.

Conclusion

In 2020, many countries have conducted statistical analysis on tuberculosis. According to world TB report, 78% of the TB population is affected by MDR-TB, and the count for new cases is on the rise. Though PZA resistance is alarmingly high among TB patients, it is still a matter of debate to include it in standard and novel drug regimen as routine PZA resistance testing is required; it is done with the help of phenotypic methods which require several weeks to confirm the result leading in the delay and onset of treatment. WHO has included PZA in MDR-TB treatment, but the conventional method including BACTEC MGIT 960, which is generally used to detect PZA resistance, sometimes provides false positive result, leading to misdiagnosis.

There is no approved rapid test for PZA resistance other than Geno scholar PZA TB II which is a hybridization assay developed by Nipro company. It can reduce the delay in initiating treatment for TB and DR-TB cases by providing results much faster than general methods, but it has not yet been approved by WHO. With advancement in sequencing technologies, the most reliable technique to detect PZA resistance among clinical samples includes genotypic methods which are very rapid and can give results in hours compared to DST which takes several weeks. Next-generation sequencing is more reliable and faster. It is extremely parallel, sequencing millions of fragments, contributing greater discovery in detection of novel and rare variants by deep sequencing as it has been noted the PZA resistance can take place due to a significant mutation in the *pncA* gene or any other gene which can be detected only through whole genome sequencing.

This review provides a broad insight about the mutations responsible for the development in PZA resistance and, mechanism of action behind it. It also shares the information about different techniques to detect the resistance prioritizing molecular detection over conventional phenotypic methods, as the genotypic methods are more rapid and reliable compared to the latter. In high TB burden countries, very limited studies are only available on the detection of PZA resistance and characterization of *pncA* mutation. Moreover, there is a lack of evidence for new mechanism of genes associated with PZA resistance and genetic diversity in *M. tuberculosis*. Hence, there is a vital need to study the prevalence of PZA resistance to find out emerging unknown mechanism of resistance among global TB population.

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Declarations

Conflict of interest Authors declare no conflict of interest.

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